Big Data Solution for Warsaw Public Transit delay prediction - Milestone 1 report

Big Data Analytics

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1 Description

The goal of the project is to build a system that shows current delays in Warsaw's public transport, how they will change in the near future, and how these delays depend on weather and air conditions. The system will collect live positions of buses and trams, timetables, weather data and air quality data published by the city. By comparing the real position of a vehicle with its timetable we can calculate its delay. Then this information is combined with current weather and used both for live monitoring and for training a machine learning model that predicts delays.

Weather is an obvious candidate for a factor that affects delays, because rain, snow or ice make traffic slower. Air quality is less obvious, but we expect some correlation with delays, since heavy traffic both slows down buses and makes the air quality worse. One of the goals of the project is to check how strong these relations really are.

From the user perspective, the final product is a web dashboard which presents current delays, predicted delays and historical statistics in a simple form.

2 Project Deliverables and User Benefits

2.1 Target users

We identified three main groups of users of the system:

- **Passenger** - a person who travels by bus or tram in Warsaw and wants to know if their line is on time and what delay to expect, especially in bad weather.
- **Dispatcher** - a person working for the transport operator who watches the whole network during the day and needs to react quickly when delays start to grow.
- **Transport planner / analyst** - a person who works on timetables and the transport network in the long term and wants to understand where and why delays appear.

2.2 Deliverables

After finishing the project, the users will get access to the following features:

- **Live delay monitoring** - current position and delay of vehicles, which can be filtered by line or stop, and a map of the network with delays marked on it. Next to the delay information, the dashboard also shows current weather and air quality measured by stations located in the city.
- **Delay prediction** - predicted delay of a given line at a given stop for the near future, calculated by a machine learning model working on streaming data and taking into account current weather and air quality. Based on the predictions, the dashboard also shows alerts with a list of lines and areas where the predicted delay is higher than a chosen threshold, so problems can be noticed beforehand.
- **Historical statistics and weather impact analysis** - average delays by line, stop, hour of the day, day of the week and weather conditions, calculated with batch processing on all collected data. This also includes a comparison of delays in different weather conditions (for example dry vs. rainy days) and a list of lines and stops which are most sensitive to bad weather.

2.3 Use cases

The use case diagram of the system is presented in Figure 1, and short descriptions of each use case are given in Table 1.

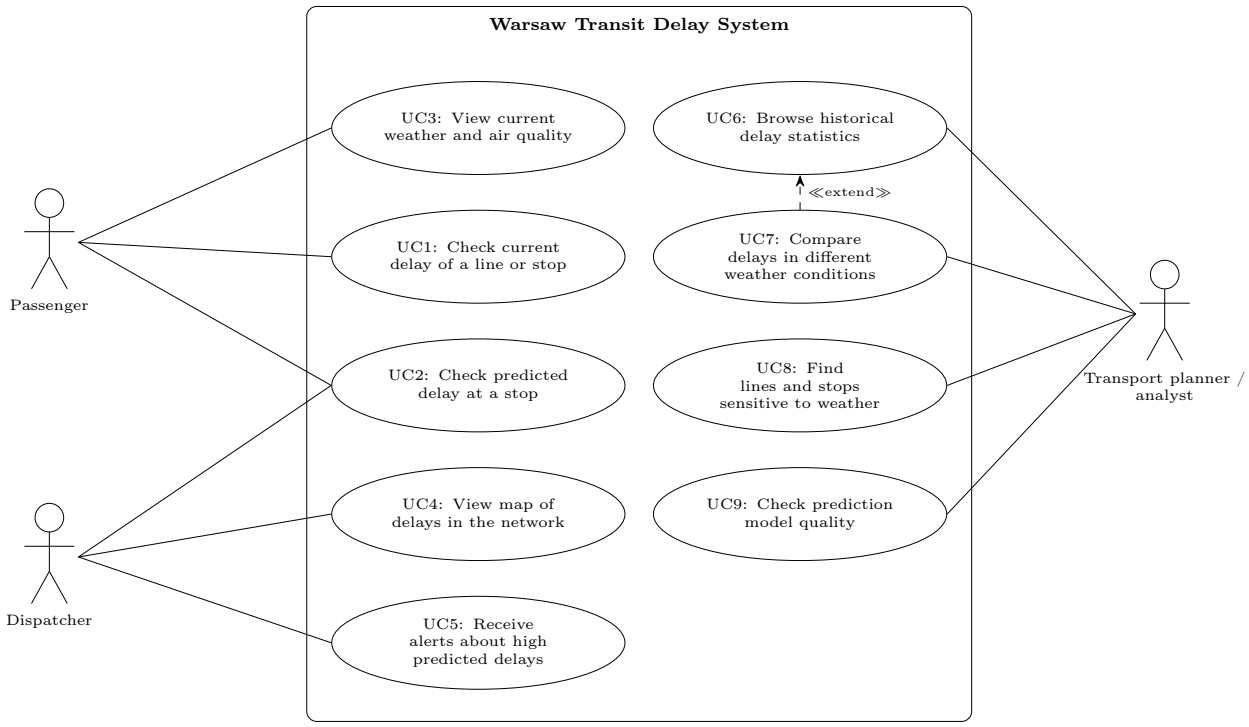


Figure 1: Use case diagram of the system

Table 1: Description of use cases

ID	Name	Actor	Description
UC1	Check current delay of a line or stop	Passenger	The user selects a line or a stop and sees the current delay of vehicles, calculated from live GPS data and the timetable.
UC2	Check predicted delay at a stop	Passenger, Dispatcher	The user selects a line and a stop and gets a predicted delay for the near future, generated by the ML model using live transit and weather data.
UC3	View current weather and air quality	Passenger	The user sees current temperature, precipitation, wind, visibility and air quality in different parts of the city.
UC4	View map of delays in the network	Dispatcher	The user sees a map with current vehicle positions coloured by their delay, which helps to quickly find problematic areas.
UC5	Receive alerts about high predicted delays	Dispatcher	The dashboard shows a list of lines where the predicted delay is above a chosen threshold.
UC6	Browse historical delay statistics	Planner	The user browses average delays by line, stop, hour and day of the week, calculated in batch mode on all collected data.
UC7	Compare delays in different weather conditions	Planner	An extension of UC6, where the user filters the statistics by weather conditions (e.g. rain, snow, low temperature) and compares the results.
UC8	Find lines and stops sensitive to weather	Planner	The user gets a ranking of lines and stops where delays grow the most in bad weather.
UC9	Check prediction model quality	Planner	The user sees evaluation metrics of the model (e.g. mean absolute error of predicted delay) and how they change over time.

2.4 Benefits for the users

Passengers will be able to plan their trips better. Instead of only seeing the timetable, they will know the current and the expected delay of their line, and they will see when bad weather is likely to make travel longer. This can help them decide to leave home earlier or choose a different line.

Dispatchers will get one place where they can see the whole network, current delays and predicted problems. Thanks to the predictions and alerts they can react earlier, for example by sending an additional vehicle or informing passengers, instead of reacting only after the delay has already grown.

Transport planners and analysts will get access to historical data that is already cleaned and combined with weather data. This makes it possible to find lines and stops which are regularly late or which are especially affected by bad weather, and to use this knowledge when planning timetables, for example by adding extra time on some routes during winter.

3 Datasources

Data used in the project will be obtained from data published by the city of Warsaw through its open data APIs, available at (`https://dane.um.warszawa.pl`), but with multiple endpoints of both streaming and batch nature. Polled data is returned in json format for all endpoints available. All data size estimations were acquired using models available in Gemini free tier.

3.1 Streaming data

- **Vehicle position data** - live GPS pings for every active vehicle: `Lat`, `Lon`, `Time`, `Lines`, `Brigade`, `VehicleNumber`. During rush hour there are about 450 trams and 1400 buses deployed, and such information is refreshed every 10s, though we would probably poll it in larger intervals, in order not to overload our limited infrastructure and not to exceed potential rate-limiting. A result of single poll has estimated size of 200KB.

- **Weather data** - air and surface temperature, precipitation, humidity, wind, and visibility for 18 stations located throughout the city. This data is refreshed every 10 minutes, though in order to use it as a streaming data source it is possible to poll it in shorter intervals for this purpose. A result of a single poll is estimated size of 4KB.

- **Air quality data** - returns a json with short description, monitoring station parameters, including type and location, and air quality parameters in form of nested records for 108 stations located throughout the city. This data is refreshed every 10 minutes, though in order to use it as a streaming data source it is possible to poll it in shorter intervals for this purpose. A result of a single poll is estimated size of 120KB.

3.2 Batch data

- **Timetable data** (static data, which should be acquired once per day) - stop locations, line-stop assignments, and per-brigade precise schedules for every stop-line-brigade combination in the network. This data daily (assuming daily batches) estimated size is at 140-200MB.

4 Data Processing Steps

The proposed solution is based on the **Lambda Architecture**, which combines the processing of current observations with the analysis of historical data. These two paths support real-time delay predictions and historical statistics, respectively. Their results are brought together in a serving layer and displayed through a dashboard. Figure 2 presents the overall data flow.

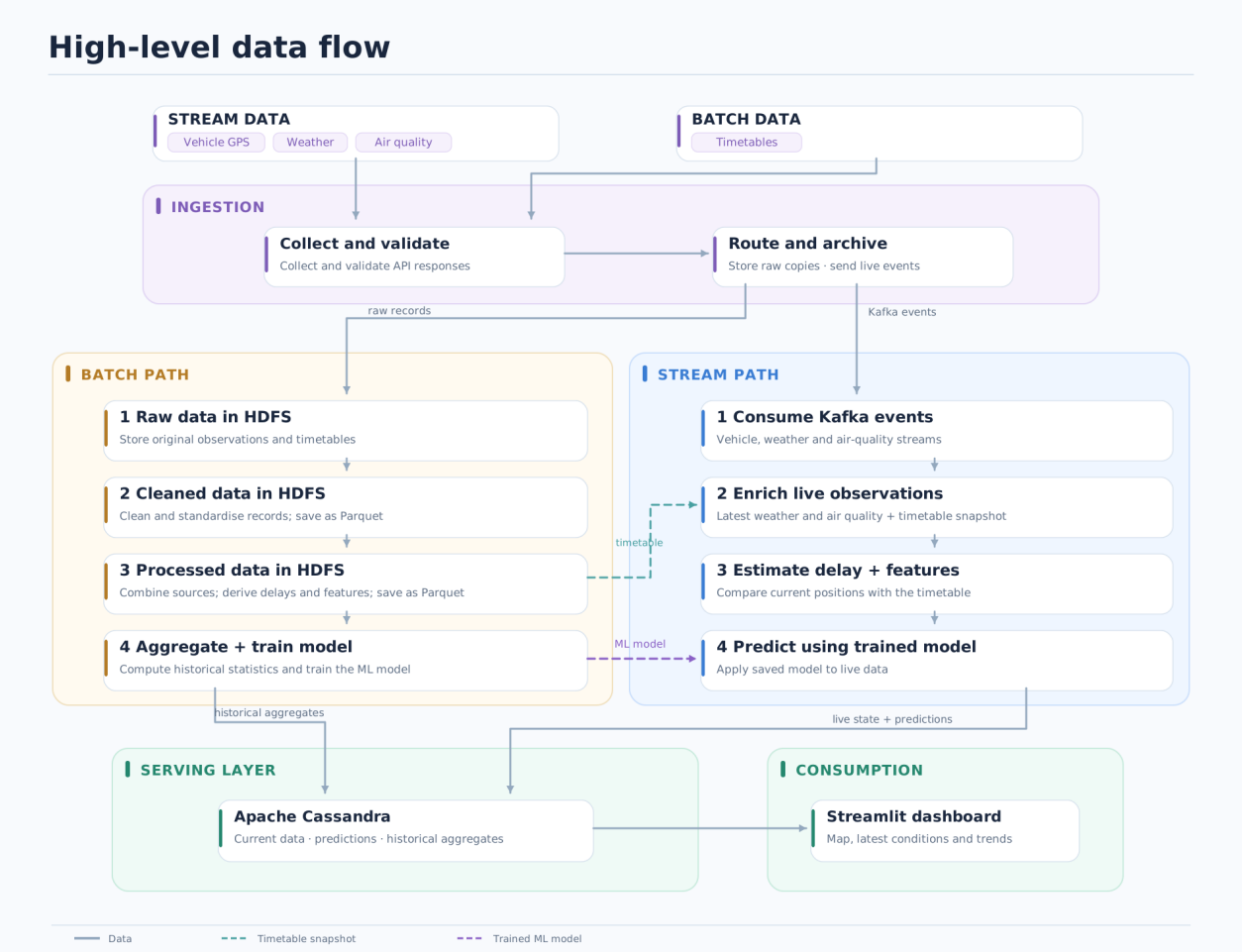


Figure 2: High-level data flow for stream and batch processing.

4.1 Data ingestion

The system will collect three types of stream data: vehicle GPS positions, weather measurements, and air quality measurements. Public transport timetables will form a separate batch source, refreshed approximately once per day. The ingestion component will retrieve and validate data from the corresponding APIs. Original responses from all four sources will be archived in HDFS, while incoming GPS and environmental observations will also be forwarded to Kafka for stream processing.

4.2 Batch processing

The batch path will use the historical data stored in HDFS. Processing will produce three groups of datasets: **Raw**, containing the original source records; **Cleaned**, containing standardised and validated data; and **Processed**, containing integrated data and analytical results. The cleaned and processed datasets will be stored in Parquet format.

Historical GPS observations will be combined with timetables and environmental measurements to estimate delays and prepare features for machine learning. The batch path will also calculate statistics such as average delays by line, stop, time period, and weather conditions. These datasets will remain available in HDFS, while selected aggregates will be transferred to the serving layer.

A prediction model will be trained and evaluated using the processed historical data. The resulting model will be saved in HDFS for use by the stream path. The batch path will also make the latest processed timetable available for real-time calculations.

4.3 Stream processing

The stream path will continuously receive GPS, weather, and air quality events from Kafka. Current vehicle observations will be combined with the latest environmental information and the processed timetable to determine the current transport situation. The stream processor will then prepare input features and use the previously trained model to predict delays.

The resulting vehicle positions, environmental measurements, estimated delays, predictions, and current statistics will be written to Cassandra. This provides the serving layer with regularly updated information about public transport conditions.

4.4 Data consumption

The dashboard will combine current information from the stream path with historical aggregates generated by the batch path. Users will be able to view vehicle positions, environmental conditions, historical delay patterns, and delay predictions in one application.

5 Solution Infrastructure

The system will use open-source Big Data technologies organised into ingestion, batch, speed, serving, and consumption components. Figure 3 shows the proposed technologies and their main connections.

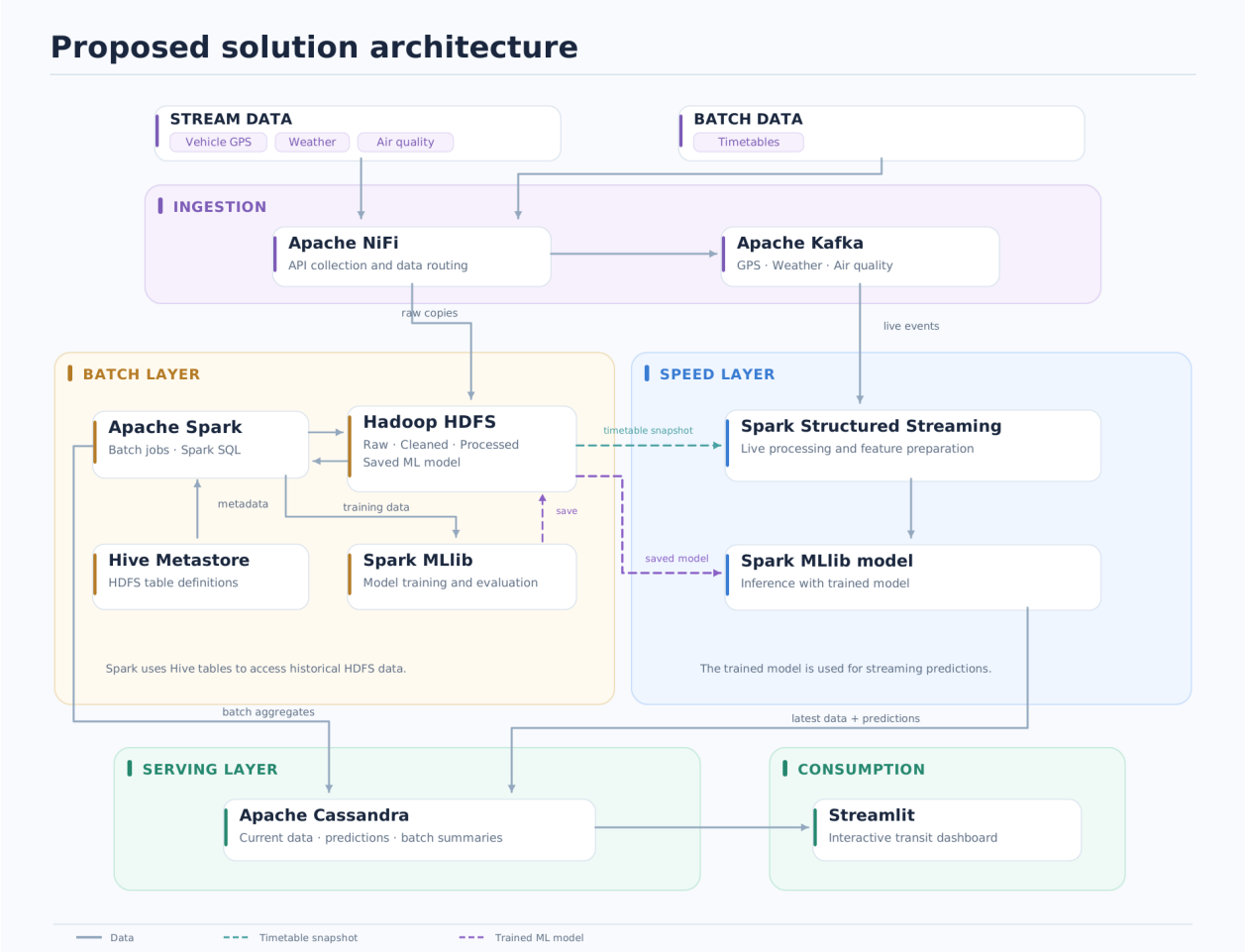


Figure 3: Proposed technology architecture based on the Lambda model.

5.1 Ingestion infrastructure

Apache NiFi will coordinate data collection from the external APIs. It will store raw copies of the collected observations and timetables in HDFS for historical processing, while forwarding live observations to **Apache Kafka**. Kafka will provide separate topics for GPS, weather, and air quality data.

5.2 Batch layer infrastructure

Hadoop HDFS will be the main historical storage system, holding Raw, Cleaned, and Processed datasets, timetable snapshots, and trained models. **Apache Spark** will perform batch transformations, source integration, and aggregation. **Hive Metastore** will store table definitions for the cleaned and processed files in HDFS, allowing Spark SQL to access the datasets as structured tables. **Spark MLlib** will be used to train and evaluate the prediction model using historical features prepared by Spark. The trained model will be stored in HDFS and made available to the speed layer.

5.3 Speed layer infrastructure

Spark Structured Streaming will consume events from Kafka and process current observations. It will enrich them with the timetable prepared by the batch layer, calculate current features, and generate predictions using the saved **Spark MLlib** model. The predictions and other current results will be written to Cassandra.

5.4 Serving and dashboard infrastructure

Apache Cassandra will store current data and predictions from the speed layer together with selected historical aggregates and model evaluation metrics from the batch layer. **Streamlit** will provide the interactive dashboard and retrieve the information needed for its visualisations from Cassandra.

5.5 Deployment

The initial solution will be deployed using **Docker Compose**, with the main technologies running as containerised services. This arrangement will simplify development and integration of the individual project components.

6 Use of Artificial Intelligence

AI tools will be used throughout the project to support software development, technical problem-solving, and report preparation. Their use will focus on improving productivity and assisting the team while ensuring that **all generated content and code are reviewed and verified by the team members**.

The planned uses of AI include:

- **Code development and editing:** Using AI tools integrated into the IDE to assist with writing, editing and debugging the code.
- **Technical support:** Using AI to clarify technical concepts, explain unfamiliar technologies, suggest implementation approaches, and assist in troubleshooting errors.
- **Report writing and editing:** Using AI to help structure the report, improve clarity and readability, correct grammatical and linguistic errors, and refine technical descriptions.
- **Verification of results:** Reviewing and testing AI-generated code, suggestions, and written content to ensure their correctness, relevance, and consistency with the actual project implementation.

AI will serve as a supporting tool rather than a replacement for the team's own work. All team members will remain responsible for understanding, validating, and maintaining the code and documentation they produce.

7 Division of Responsibilities

The project work is divided among four team members, each responsible for a specific component of the system and supported by another team member.

7.1 Data Ingestion and Validation

Main responsible person: Michał, supporting team member: Sebastian.

- Selecting and testing the weather API and public transport data source.
- Creating scripts in order to retrieve data from external APIs.
- Configuring Apache NiFi to automate the data ingestion process.
- Configuring Apache Kafka topics, such as `gps`, `weather` and `air_quality`.
- Implementing data validation, error handling and duplicate detection.
- Defining and documenting data schemas.
- Documenting the data collection process.

Expected outcome: A working data ingestion pipeline that retrieves, validates, and delivers data to Apache Kafka.

7.2 Data Storage and Batch Processing

Main responsible person: Emil, supporting team member: Michał.

- Configuring Apache Hadoop, HDFS and Apache Hive.
- Designing a directory structure for raw, cleaned, and processed data.
- Creating Hive tables and developing SQL queries.
- Cleaning the collected data and handling missing or invalid values.
- Combining weather and public transport data into a unified dataset.
- Implementing batch processing jobs using Apache Spark.
- Calculating historical statistics, such as average bus delays by route and weather conditions.

Expected outcome: A functional data storage layer, a unified dataset, and a repeatable batch processing pipeline for historical analysis.

7.3 Stream Processing and Machine Learning

Main responsible person: Tymoteusz, supporting team member: Emil.

- Configuring Apache Spark Structured Streaming.
- Implementing the consumption of events from Apache Kafka.
- Calculating real-time metrics and statistics using streaming data.
- Preparing features from weather and public transport data for machine learning.
- Training a machine learning model to predict bus delays.
- Evaluating model performance with metrics.
- Integrating the trained model into the streaming pipeline to generate predictions.

Expected outcome: A working stream processing pipeline capable of calculating real-time metrics and generating bus delay predictions using a trained ML model.

7.4 Results Serving, Dashboard, and System Integration

Main responsible person: Sebastian, supporting team member: Tymoteusz.

- Configuring a database and creating tables for analytical results and predictions.
- Defining common data formats and interfaces with the other team members.
- Integrating the results of batch and stream processing.
- Developing an interactive dashboard using Streamlit.
- Visualising weather conditions, bus delays, historical statistics, and machine learning predictions.
- Integrating system components using Docker Compose.
- Testing the overall system and resolving integration issues.
- Preparing instructions for setting up and running the complete system.

Expected outcome: A functional dashboard presenting the results of the project, an integrated system, and documentation explaining how to run the application.